L4: Power-law Degree Distribution

A ***"power-law network"*** has a degree distribution that is defined by the following equation:

$$p_k = c\, k^{-\alpha}$$

In other words, the probability that the degree of a node is equal to a positive integer k is proportional to $k^{-\alpha}$ where α >0.

The proportionality coefficient **c** is calculated so that the sum of all degree probabilities is equal to one for a given α and a given minimum degree k_{min} (the minimum degree may not always be 1).

$$\sum_{k=k_{min}}^{\infty} p_k = 1$$

The calculation of **c** can be simplified if we approximate the discrete degree distribution with a continuous distribution:

$$\int_{k=k_{min}}^{\infty} p_k = c \int_{k=k_{min}}^{\infty} k^{-\alpha} = -\frac{c}{\alpha-1} k^{-(\alpha-1)}|_{k_{min}}^{\infty} = \frac{c}{\alpha-1} k_{min}^{-(\alpha-1)} = 1$$

which gives:

$$c = (\alpha - 1)\ k_{min}^{\alpha-1}$$

So, the complete equation for a power-law degree distribution is:

$$p_k = \frac{\alpha-1}{k_{min}}\,(\frac{k}{k_{min}})^{-\alpha}$$

The Complementary Cumulative Distribution Function (C-CDF) is:

$$P[\text{degree} \geq k] = (\frac{k}{k_{min}})^{-(\alpha-1)}$$

Note that the exponent of the CCDF function is $\alpha - 1$, instead of α. So, if the probability that a node has degree k decays with a power-law exponent of 3, the probability that we see nodes with degree greater than k decays with an exponent of 2.

For directed networks, we can have that the in-degree or the out-degree or both follow a power-law distribution *(with potentially different exponents).*

Food for Thought

Prompt 1: Repeat the derivations given here in more detail.

Prompt 2: Can you think of a network with n nodes in which all nodes have about the same in-degree but the out-degrees are highly skewed?